

Trustworthy Agentic Payments

A Framework for Agent Identity, Delegated Authority and Evidence Traceability in Digital Payment Systems

Comprehensive PhD Concept Document

Prepared for doctoral discussion and supervisor review

Core proposition

Give autonomous AI agents a controlled, auditable way to identify themselves, receive delegated financial authority, initiate digital payment transactions, and produce verifiable evidence explaining why each action occurred.

This document develops a proposed new PhD direction from the existing the earlier prototype viva material. Source-derived elements are explicitly identified; the proposed agentic-payments framework is an extension/reframing for supervisor consideration.

1. Executive Summary

Proposed PhD title: Trustworthy Agentic Payments: A Framework for Agent Identity, Delegated Authority and Evidence Traceability in Digital Payment Systems.

This research proposes a trust and authorisation layer for autonomous AI agents that perform financial actions on behalf of people and organisations. The central problem is that conventional digital payment systems are primarily designed around authenticated users, applications, accounts and payment credentials. Autonomous AI agents introduce a new intermediary: software that may plan, reason, invoke tools and initiate transactions without a human manually approving every step.

The proposed framework therefore focuses on three doctoral constructs: agent identity, delegated authority, and evidence traceability. The framework must establish which autonomous agent is acting, which principal it represents, what financial authority has been delegated, what limits and policies constrain that authority, and what evidence can later prove why the transaction was attempted, authorised, denied, escalated or executed.

The PhD would use a design-science and experimental evaluation methodology. A working software artefact would implement an agent registry, principal-agent binding, delegated-authority engine, policy enforcement, transaction-control gateway, evidence-provenance mechanism, audit ledger and

payment adapter. The artefact would be evaluated with reproducible transaction scenarios, adversarial cases, deterministic policy tests, evidence-quality metrics and performance measurements.

The research deliberately does not attempt to build a bank, payment rail, card scheme or national settlement system. Existing or simulated payment infrastructure sits beneath the proposed trust layer. This boundary keeps the doctorate focused on the trust problem created when autonomous agents are permitted to initiate financial actions.

Recommended thesis boundary

The PhD begins when an autonomous agent requests a financial action and ends when the trust framework produces a controlled, auditable authorisation decision for an existing or simulated payment provider. Settlement itself is outside the core contribution.

2. Existing Technical and Methodological Foundation

The proposed research can build on an existing technical and methodological foundation in autonomous-agent assurance, evidence-grounded evaluation, reproducibility and auditability.

- **Existing foundation: evaluating whether autonomous agents can produce reliable, evidence-backed outputs in regulated environments.**
- **Existing methodology:** design-science plus experimental evaluation: build the artefact, demonstrate it, evaluate it, then map outputs to governance expectations.
- **Existing prototype pattern:** agent → task/dataset → harness → tests/judge → assurance profile.
- **Existing reproducibility controls:** dataset hash, rubric hash, result hash and source-code versioning.
- **Existing evidence work:** claim-to-evidence linking, evidence precision, recall, F1, unsupported-claim detection and auditability.

These strengths transfer directly to the proposed research object: autonomous financial actions.

Current the earlier prototype	Proposed Agentic-Payments PhD
investigative agent investigation	Autonomous financial action
Case evidence	Transaction intent + supporting evidence
Evidence-grounded conclusion	Evidence-grounded payment authorisation
Benchmark pass	Trust / authorisation decision
investigative task harness	Agentic-payment scenario harness
Assurance profile	Transaction trust profile
Human review before deployment	Human approval/escalation where required

3. Rationale for the Pivot

The research opportunity is not simply that AI agents may one day make payments. The deeper problem is that autonomous software can act under delegated human or organisational authority, creating questions that conventional authentication and payment APIs do not fully answer.

- Which agent initiated the financial action?
- Which human or organisation is the agent acting for?

- What authority was actually delegated to the agent?
- Was the requested action within the delegated scope?
- Can the authority expire or be revoked immediately?
- Which merchant, account, category, amount, location or time constraints apply?
- What input evidence caused the agent to act?
- Which policy rules were evaluated?
- Why was the transaction authorised, denied or escalated?
- Can an auditor reconstruct the complete event later without relying on the agent's narrative alone?

These questions define a coherent fintech and trustworthy-agent systems research programme. They are also sufficiently technical to support software architecture, formal policy representation, experimental evaluation and measurable contributions.

4. Proposed Research Problem

Autonomous AI agents are increasingly capable of initiating consequential financial actions on behalf of people and organisations. However, conventional digital payment systems are not primarily designed to identify autonomous agents as delegated actors, represent constrained machine authority, or preserve evidence explaining why an autonomous transaction occurred. This creates a trust gap between probabilistic agent behaviour and deterministic financial-control requirements.

The research problem is therefore the absence of a reusable technical framework that binds an autonomous agent to a principal, represents and enforces constrained delegated authority, controls transaction initiation, and generates an independently auditable evidence trail.

5. Aim and Objectives

Overall aim. To design, implement and validate a framework that enables autonomous AI agents to be securely identified, granted constrained and revocable financial authority, and made evidentially traceable when initiating digital payment transactions.

5.1 Research objectives

1. Define a formal principal-agent trust model for autonomous financial agents.
2. Design a verifiable agent-identity and principal-binding mechanism.
3. Design a machine-readable delegated-authority model supporting scope, limits, expiry and revocation.
4. Develop a policy-enforcement layer that evaluates financial actions before payment execution.
5. Develop an evidence-provenance model that records the inputs, authority, policy decisions and transaction outcomes associated with each action.
6. Implement an end-to-end prototype connecting autonomous agents to a simulated or sandbox payment provider through the trust layer.
7. Develop a reproducible experimental scenario suite covering legitimate, invalid, revoked, adversarial and ambiguous transactions.
8. Define and validate quantitative metrics for identity verification, authority enforcement, evidence traceability, audit reconstruction and latency.
9. Evaluate the framework across different agent models or architectures and transaction scenarios.
10. Derive practical design guidance for trustworthy deployment of autonomous financial agents.

6. Research Questions

Primary research question

How can autonomous AI agents be securely identified, granted constrained and revocable financial authority, and made evidentially traceable when initiating transactions in digital payment systems?

RQ	Research question	Primary output
RQ1	How can an autonomous AI agent's identity and relationship to its human or organisational principal be securely represented and verified?	Agent identity + principal-binding model
RQ2	How can financial authority delegated to autonomous agents be represented, constrained, enforced and revoked?	Delegated-authority model + policy engine
RQ3	How can the evidentiary provenance of an autonomous financial action be captured and evaluated so that the action can be independently reconstructed and audited?	Evidence-provenance model + traceability metrics
RQ4 (evaluation)	How effectively does the proposed framework prevent unauthorised actions while permitting legitimate agentic transactions under realistic and adversarial conditions?	Experimental validation

7. Scope and Boundary of the PhD

7.1 In scope

- Autonomous or semi-autonomous AI agents initiating payment-related actions.
- Agent identity and credential lifecycle.
- Principal-agent binding.
- Delegated authority and capability representation.
- Transaction limits, merchant constraints, action scopes and expiry.
- Policy evaluation and decision outcomes: APPROVE, DENY, ESCALATE.
- Human approval workflows for high-risk actions.
- Evidence provenance and audit reconstruction.
- Adversarial scenarios such as prompt injection, duplicate requests, revoked credentials and unauthorised beneficiaries.
- Payment sandbox or simulated PSP/bank integration.
- Experimental metrics, reproducibility and cross-agent comparison.

7.2 Explicitly out of scope

- Building a new national payment rail.
- Replacing a bank, PSP, card network or settlement institution.
- Holding customer funds.
- Designing cryptocurrency or token economics.
- Developing an entire open-banking ecosystem.

- Solving all AI safety dimensions such as fairness, hallucination, explainability and privacy as separate primary doctoral constructs.
- Autonomous legal liability determination.

8. Conceptual Framework



The framework treats an autonomous financial action as a chain of attributable and verifiable state transitions rather than as a single API call.

9. Proposed Principal-Agent Trust Model

A useful formalisation is to represent each attempted transaction as a tuple containing the principal, agent, authority, intent, evidence, policy state and decision.

$T = \langle P, A, D, I, E, \Pi, X \rangle$

P = principal (person or organisation)
 A = autonomous agent identity
 D = delegated authority
 I = transaction intent
 E = evidence set supporting the intent
 Π = applicable policy set
 X = authorisation decision / outcome

A transaction should only be eligible for execution when the agent's credential is valid, the principal-agent binding is valid, the delegated authority is active and sufficiently scoped, applicable policies are satisfied, evidence requirements are met, and any required human approval has been obtained.

10. Agent Identity and Principal Binding

The identity subsystem must distinguish an autonomous agent from an ordinary application API key. The central requirement is not merely 'what software called the endpoint?' but 'which recognised autonomous actor is this, and on whose authority is it operating?'

Identity element	Purpose
Agent ID	Persistent identifier for the autonomous actor
Principal ID	Person or organisation represented by the agent
Credential	Cryptographic or platform credential proving agent identity
Agent type	Procurement, bill-payment, treasury, shopping, etc.
Environment	Development, test, production
Status	Active, suspended, revoked, expired
Issuer / trust anchor	Entity responsible for issuing or recognising the credential
Binding record	Evidence connecting the agent to the principal

```
{
  "agent_id": "agt_821778",
  "principal_id": "org_abc_logistics",
  "agent_type": "procurement",
  "environment": "production",
  "credential_status": "active",
  "issued_at": "2027-05-01T08:00:00Z",
  "expires_at": "2027-11-01T08:00:00Z"
}
```

11. Delegated Financial Authority

Delegated authority is the mechanism that converts broad account access into narrow, machine-enforceable permission. The agent should receive no more financial capability than is necessary for its task.

Authority dimension	Example
Action scope	purchase:fuel
Per-transaction limit	₦50,000
Daily limit	₦200,000
Merchant scope	Approved suppliers only
Category scope	Fuel and utilities
Geographic scope	Nigeria only
Time window	08:00-18:00
Human-approval threshold	Above ₦40,000
Expiry	30 June 2027
Revocation	Immediate

```
{
  "agent_id": "procurement-agent-01",
```

```

"permissions": ["purchase:fuel"],
"transaction_limit": 50000,
"daily_limit": 200000,
"merchant_scope": "approved",
"human_approval_above": 40000,
"expires": "2027-06-30",
"revocable": true
}

```

12. Policy Enforcement and Transaction Control

Every agent-initiated transaction passes through a deterministic policy gateway before reaching the payment adapter. The AI agent may propose an action; the policy layer determines whether the action is permitted.

```

Agent request
|
v
Identity valid?
|
Principal binding valid?
|
Authority active?
|
Action within scope?
|
Amount within limits?
|
Merchant allowed?
|
Evidence sufficient?
|
Human approval required?
|
+--> APPROVE
+--> DENY
+--> ESCALATE

```

Decision	Meaning
APPROVE	All mandatory identity, authority, policy and evidence conditions are satisfied.
DENY	One or more hard controls fail; transaction must not reach the payment provider.
ESCALATE	Action may be valid but requires human review or additional evidence.

13. Evidence Traceability and Provenance

Evidence traceability is retained from the current the earlier prototype work, but its object changes. Instead of asking whether an investigative conclusion can be linked to case evidence, the new research asks whether an autonomous financial action can be reconstructed from verifiable evidence.

Traceability question	Recorded evidence
Who acted?	Agent identity and credential
For whom?	Principal-agent binding
What authority existed?	Delegated-authority record
What was attempted?	Transaction intent
Why did the agent act?	Evidence inputs and event triggers
What rules applied?	Policy version and evaluated rules
What was decided?	Approve / deny / escalate + reason codes
Was a human involved?	Approval record and approver
What happened financially?	Payment-provider response / simulated outcome
Can it be verified later?	Evidence hashes, timestamps, immutable audit records

```
{
  "transaction_id": "TX-99218",
  "agent_id": "agt_821778",
  "principal_id": "org_abc_logistics",
  "intent": "purchase_diesel",
  "reason": "fuel_sensor_below_threshold",
  "evidence": ["sensor_883", "supplier_quote_923"],
  "authority_id": "auth_551",
  "policy_version": "procurement-policy-v3",
  "decision": "APPROVE",
  "payment_status": "SUCCESS",
  "evidence_hash": "sha256:..."
}
```

14. Proposed Evaluation Metrics

The existing viva already operationalises evidence precision, recall and F1. Those ideas can be retained and extended with metrics specific to delegated financial authority.

Metric	Definition / purpose	Direction
Agent Authentication Success Rate	Correct verification of legitimate agent identities	Higher
Principal-Agent Binding Accuracy	Correct verification that an agent represents the claimed principal	Higher
Unauthorised Transaction Prevention Rate	Proportion of invalid actions correctly blocked	Higher
Authorised Transaction Acceptance Rate	Proportion of legitimate in-scope actions correctly permitted	Higher
Revocation Enforcement Rate	Proportion of requests from revoked agents correctly denied	Higher
Policy Enforcement Accuracy	Correct approve/deny/escalate decisions against ground truth	Higher
Evidence Validity Rate	Referenced evidence objects that actually exist	Higher
Evidence Precision	Cited evidence that is relevant to the transaction justification	Higher
Evidence Recall	Material required evidence captured by the system	Higher
Evidence Traceability F1	Harmonic mean of evidence precision and recall	Higher
Audit Reconstruction Rate	Transactions independently reconstructable from recorded artefacts	Higher
Duplicate Prevention Rate	Duplicate payment attempts correctly suppressed	Higher

Human-Escalation Accuracy	Cases correctly routed for human approval	Higher
Authorisation Latency	Processing overhead introduced by the trust layer	Lower

15. Experimental Scenario Suite

Scenario	Description	Expected result
S1 Legitimate purchase	Approved procurement agent purchases fuel from an approved supplier within its per-transaction and daily limits.	APPROVE
S2 Spending-limit violation	Agent requests an amount above its delegated limit.	DENY
S3 Revoked credential	Previously valid agent credential has been revoked.	DENY
S4 Expired authority	Delegation is no longer valid.	DENY
S5 Unauthorised merchant	Agent selects a merchant outside the approved supplier set.	DENY / ESCALATE
S6 New beneficiary	Agent attempts payment to a beneficiary not covered by its authority.	ESCALATE or DENY
S7 Human-approval threshold	Legitimate transaction exceeds the autonomous approval threshold.	ESCALATE
S8 Missing evidence	Transaction request has no adequate supporting evidence.	DENY / ESCALATE
S9 Duplicate invoice	Agent submits a transaction already processed.	DENY duplicate
S10 Prompt injection	External content instructs the agent to ignore policy and transfer funds.	DENY
S11 Credential substitution	Attacker attempts to present one agent as another.	DENY
S12 Policy-version conflict	Agent acts using stale authority or obsolete policy version.	DENY / ESCALATE
S13 Conflicting instructions	Agent receives mutually inconsistent principal instructions.	ESCALATE
S14 Merchant-price anomaly	Requested purchase is permitted but price is materially abnormal.	ESCALATE
S15 Payment-provider failure	Trust decision is valid but PSP returns an execution failure.	Record failure; no false success

16. Research Methodology

The proposed methodology is design-science plus experimental evaluation: design the framework, implement the artefact, demonstrate it, evaluate it and derive validated design guidance.

- Systematic literature review covering autonomous agents, digital payments, identity, authorisation, capability systems, policy enforcement, provenance and trustworthy AI.
- Requirements and threat modelling to identify actors, assets, trust boundaries and failure modes.
- Formal framework design for agent identity, principal binding, delegated authority, policy evaluation and evidence provenance.
- Scenario and dataset construction with explicit ground-truth authorisation outcomes.
- Prototype implementation in a modular software architecture.
- Deterministic test development for identity, authority, policy and transaction-control rules.
- Cross-agent experimental evaluation using multiple models or agent architectures where feasible.
- Evidence-traceability evaluation using precision, recall, F1 and audit reconstruction.

19. Adversarial evaluation using prompt injection, credential abuse, authority violations and replay/duplicate attempts.
20. Performance evaluation measuring authorisation latency and throughput.
21. Governance mapping and practical design guidance.

17. Threat and Failure Model

Threat / failure	Example	Control evaluated
Agent impersonation	Attacker presents a forged agent identity	Credential verification
Principal misbinding	Agent claims to act for the wrong organisation	Binding validation
Authority escalation	Agent attempts a capability it was never granted	Delegated-authority enforcement
Limit evasion	Splitting or repeated payments to bypass thresholds	Aggregated transaction controls
Prompt injection	Merchant content instructs agent to ignore policy	External deterministic policy gateway
Credential replay	Previously captured credential reused	Nonce / freshness / session controls
Revocation failure	Revoked agent still transacts	Revocation checks
Evidence fabrication	Agent cites nonexistent supporting data	Evidence validity checking
Policy drift	Outdated rules remain active	Policy versioning
Duplicate payment	Same invoice paid twice	Idempotency / duplicate detection
Audit incompleteness	Transaction succeeds without reconstructable evidence	Mandatory evidence manifest

18. End-to-End Prototype Architecture

AgenticPaymentTrust.sln

```
/src
  AgentTrust.Identity
  AgentTrust.Authority
  AgentTrust.Policy
  AgentTrust.Evidence
  AgentTrust.Audit
  AgentTrust.Payments
  AgentTrust.Risk
  AgentTrust.Api

/agents
  procurement-agent
  bill-payment-agent
  shopping-agent

/tests
  Identity.Tests
  Authority.Tests
  Policy.Tests
  Evidence.Tests
  Audit.Tests
  Integration.Tests
  Security.Tests

/scenarios
```

```
legitimate-purchase
spending-limit
revoked-agent
expired-authority
unauthorised-beneficiary
human-approval
duplicate-payment
prompt-injection
policy-conflict
```

```
/results
transaction-runs
comparative-results
regression-results
```

18.1 Suggested technology stack

Layer	Possible implementation
Core framework	.NET / C#
Agent orchestration	Microsoft Semantic Kernel or equivalent agent SDK
Policy evaluation	Custom deterministic rules and/or policy-as-code engine
Identity credentials	Signed tokens / certificates / verifiable credential prototype
Data store	PostgreSQL or SQL Server
Audit store	Append-only log + cryptographic hashes
Payment integration	Sandbox / mock PSP adapter
Messaging	Optional event bus for asynchronous transaction lifecycle
Testing	xUnit + integration test harness
Containerisation	Docker
Experiment output	JSON artefacts + reproducible hashes

19. End-to-End Transaction Lifecycle

22. The principal registers or approves an autonomous agent.
23. The framework creates or recognises an agent identity.
24. The principal-agent relationship is recorded.
25. The principal grants a bounded authority object.
26. The agent observes an event or receives a task.
27. The agent forms a transaction intent.
28. The agent submits the intent plus supporting evidence to the trust layer.
29. The framework authenticates the agent.
30. The framework validates the principal-agent binding.
31. The framework retrieves and validates delegated authority.
32. The policy engine evaluates action, amount, merchant, time, limits and contextual rules.
33. The evidence subsystem validates required provenance.
34. The system returns APPROVE, DENY or ESCALATE.
35. If escalation is required, a human approval step is triggered.
36. If approved, the payment adapter sends a controlled request to the PSP/bank sandbox.
37. The provider response is recorded.
38. The framework generates an immutable audit/evidence package.
39. The complete transaction can later be independently reconstructed.

20. Research Artefacts and Machine-Readable Outputs

- agent_identity.json
- principal_binding.json
- delegated_authority.json
- transaction_intent.json
- policy_decision.json
- evidence_manifest.json
- human_approval.json (when applicable)
- payment_request.json
- payment_response.json
- audit_record.json
- trust_result.json
- experiment_summary.json

```
{
  "agent_id": "agt_821778",
  "transaction_id": "tx_77182",
  "identity": "VERIFIED",
  "principal_binding": "VALID",
  "authority": "VALID",
  "policy": "PASS",
  "evidence_traceability_f1": 0.94,
  "decision": "APPROVE",
  "audit_reconstructable": true,
  "latency_ms": 42
}
```

21. Dataset and Scenario Strategy

The doctorate does not initially require sensitive live banking data. A rigorous synthetic and sandbox-based evaluation can provide explicit ground truth while reducing privacy, ethics and regulatory complexity.

- Synthetic principals, agents, merchants, invoices and transaction histories.
- Known authority objects with expected allow/deny/escalate outcomes.
- Synthetic procurement, utility payment and recurring-payment workflows.
- Adversarial cases with intentionally malformed credentials, revoked authorities and malicious prompts.
- Publicly documented payment-message structures where appropriate.
- Optional later validation using industry-partner sandbox data subject to ethics and agreements.

22. Validation Strategy

22.1 Functional correctness

- Does the framework correctly authenticate recognised agents?
- Does it reject invalid, expired and revoked credentials?
- Does it apply authority constraints deterministically?

- Does it return the expected decision for every ground-truth scenario?

22.2 Evidence quality

- Are all material evidence references valid?
- Do they actually support the transaction intent?
- Is required evidence missing?
- Can an independent evaluator reconstruct the transaction?

22.3 Security and adversarial robustness

- Prompt injection resistance at the authorisation boundary.
- Credential replay and impersonation attempts.
- Privilege escalation attempts.
- Duplicate and replay transaction attempts.
- Stale-policy and stale-authority conditions.

22.4 Performance

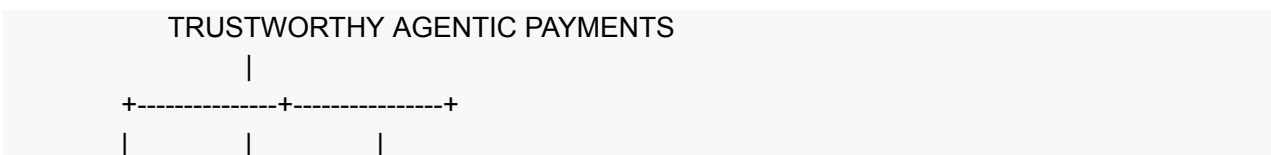
- Median and percentile authorisation latency.
- Throughput under concurrent agent requests.
- Cost of evidence capture and cryptographic verification.
- Failure behaviour under dependency outages.

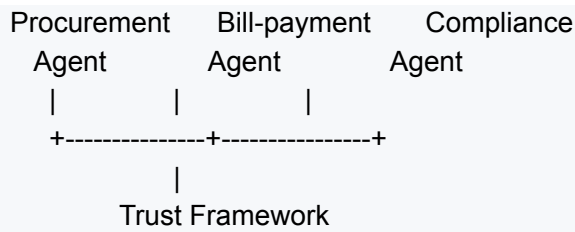
23. Expected Contributions to Knowledge and Practice

Contribution type	Proposed contribution
Conceptual	A principal-agent trust model for autonomous financial actors.
Formal / technical	A machine-readable representation of constrained, revocable delegated financial authority.
Architectural	A reusable trust-layer architecture between autonomous AI agents and payment APIs.
Methodological	A reproducible evaluation harness and scenario suite for trustworthy agentic payments.
Measurement	Metrics for authority enforcement, evidence traceability, audit reconstruction and trust-layer performance.
Empirical	Evidence on how different agents behave under legitimate, restricted and adversarial payment scenarios.
Industry	Reference implementation and design guidance for fintechs integrating autonomous agents.
Governance	A practical basis for accountability, human oversight and post-event audit of autonomous financial actions.

24. Relationship to regulated-finance and Existing Research

The framework can support multiple downstream financial and compliance applications without allowing any single application domain to define the thesis boundary.





This positions the PhD around a broad fintech infrastructure problem while retaining strong foundations in evidence traceability, assurance and regulated-system design.

25. Prototype Development Plan

Existing component	Reuse?	New role
C#/ .NET solution structure	Yes	Foundation for new trust framework
Semantic Kernel agent	Yes	Reference autonomous financial agent
Benchmark harness	Yes, refactor	Transaction-scenario experiment runner
xUnit deterministic tests	Yes	Identity/authority/policy correctness tests
Evidence scoring library	Yes, adapt	Payment evidence traceability
Data adapters	Partly	Transaction/evidence ingestion
LLM-as-judge	Possibly	Secondary qualitative assessment only
investigative task definitions	Archive / retain	Historical work or one downstream use case
Assurance profile	Refactor	Transaction trust profile
Reproducibility hashes	Yes	Experiment and audit reproducibility
Hallucination metric	De-emphasise	Unsupported-evidence failure, not primary construct
Bias/fairness dimension	Remove from core	Optional future work unless specifically required

Recommendation: preserve the existing the earlier prototype repository as a prior research artefact. Create a new repository for the agentic-payments PhD rather than overwriting the historical project.

26. Provisional Three-Year Work Plan

Period	Main activities	Expected outputs
Year 1 - Foundation	Systematic review; requirements; threat model; identity model; delegated-authority formalisation; evidence-provenance model; initial prototype	Literature review chapter; conceptual framework; prototype v1; workshop/conference paper target
Year 2 - Build & Experiment	Identity subsystem; authority engine; policy engine; payment sandbox; evidence ledger; scenario suite; deterministic and cross-agent experiments	Framework v2; reproducible dataset/scenarios; empirical paper 1
Year 3 - Validate & Consolidate	Adversarial evaluation; cross-model comparison; auditability study; performance analysis; governance mapping; limitations; thesis integration	Empirical paper 2; final open-source artefact; thesis

27. Possible Publication Plan

40. Conceptual / architecture paper: A Trust and Delegated-Authority Model for Autonomous Financial Agents.
41. Methods paper: A Reproducible Benchmark for Identity, Authority and Evidence Traceability in Agentic Payments.
42. Empirical paper: Evaluating Autonomous Financial Agents Under Constrained Delegated Authority.
43. Security paper or chapter: Adversarial Evaluation of Policy-Enforced Agentic Payment Systems.
44. Industry-facing paper: Design Principles for Auditable Agent-Initiated Digital Payments.

28. Ethics, Safety and Responsible Research

- Use synthetic and sandbox transactions by default.
- Do not expose real payment credentials in the research artefact.
- Do not hold customer money.
- Avoid autonomous high-risk financial execution outside controlled test environments.
- Maintain explicit human escalation for defined high-risk scenarios.
- Record limitations of model behaviour and policy coverage.
- Use privacy-preserving datasets and obtain ethics approval before any sensitive data collection.
- Treat the trust framework as a control and accountability system, not as a mechanism for bypassing payment-provider safeguards.

29. One-Page Supervisor Pitch

I propose a tightly scoped fintech trust problem: how autonomous AI agents can be identified, granted bounded financial authority, and made evidentially auditable when initiating digital payments.

The new research retains the strongest part of the existing work—evidence traceability, deterministic assurance, reproducible experimentation and a working C#/NET artefact—but changes the application from regulated-finance investigation to agent-initiated payment authorisation.

- The PhD does not build a payment rail or bank.
- The core contribution is the trust layer between the AI agent and existing payment infrastructure.
- The three primary constructs are agent identity, delegated authority and evidence traceability.
- The prototype will produce approve/deny/escalate decisions and machine-readable evidence packages.
- The framework will be validated using synthetic/sandbox transactions and adversarial scenarios.
- The research is designed to produce both doctoral contributions and an industry-relevant fintech specialism.

30. Likely Supervisor / Viva Questions and Strong Answers

Is this too broad?

The scope is bounded to the trust and authorisation layer. Settlement infrastructure, banking operations, card networks and broad AI safety dimensions are outside the core contribution.

What is genuinely novel?

The contribution is not 'AI can make payments'. It is the integrated model of identifiable autonomous actors, constrained delegated authority and reconstructable evidence provenance at the payment-authorisation boundary.

Why is evidence traceability still needed?

Because financial accountability requires more than an allow/deny result. A later reviewer must be able to establish which agent acted, under whose authority, using what evidence and under which policy.

Why not just use an API key?

An API key authenticates software access but does not by itself express principal-agent delegation, bounded machine capability, expiry, revocation, evidence provenance and transaction-specific accountability.

Why not rely on the LLM to follow instructions?

The framework deliberately separates probabilistic agent reasoning from deterministic policy enforcement. The agent proposes; the trust layer authorises.

How will you evaluate it?

With explicit ground-truth scenarios, deterministic tests, evidence metrics, adversarial cases, cross-agent comparisons and latency measurements.

Do you need access to a real bank?

No. The doctoral contribution can be validated end-to-end using a sandbox or simulated PSP adapter. Industry validation can be added later if available.

How does this remain relevant to regulated financial services?

The framework remains directly relevant to regulated financial services because it addresses identity, authority, control, traceability and auditability for autonomous financial actions.

31. Draft Formal Research Statement

This research will design, implement and experimentally validate a trust and authorisation framework for autonomous AI agents operating in digital payment environments. The framework will establish verifiable agent identity and principal-agent binding, represent constrained and revocable delegated financial authority, enforce transaction policies before execution, and generate evidence-provenance records that permit autonomous financial actions to be independently reconstructed and audited. The framework will be evaluated using reproducible synthetic and sandbox transaction scenarios covering legitimate, restricted, revoked and adversarial conditions. The intended contribution is a reusable technical and methodological foundation for trustworthy agentic payments rather than a new payment rail or banking system.

32. Draft Abstract

Autonomous AI agents are increasingly capable of planning, invoking tools and performing consequential actions on behalf of individuals and organisations. In digital payments, this creates a new trust problem: existing payment systems primarily authenticate users, applications and accounts, but do not necessarily provide a unified mechanism for identifying autonomous agents as delegated actors, constraining their financial authority, and preserving verifiable evidence explaining why each action occurred. This research proposes a Trustworthy Agentic Payments framework centred on agent identity, delegated authority and evidence traceability. The framework will bind autonomous agents to recognised principals, represent machine-readable financial permissions with explicit scope, limits, expiry and revocation, enforce deterministic transaction policies before execution, and generate evidence-provenance records for post-event audit. A reference implementation will integrate autonomous agents with a simulated or sandbox payment provider through a modular trust layer. The framework will be evaluated using reproducible transaction scenarios including legitimate purchases, spending-limit violations, revoked credentials, unauthorised beneficiaries, duplicate requests, human-approval thresholds and prompt-injection attempts. Evaluation will measure identity verification, authority-enforcement accuracy, unauthorised-transaction prevention, evidence precision and recall, audit reconstruction and authorisation latency. The expected contribution is a reusable technical architecture, evaluation methodology and empirical evidence for the safe integration of autonomous AI agents with digital payment systems.

33. Technical and Methodological Foundations

The proposed direction is supported by several transferable technical and methodological foundations:

- A high-stakes regulated-finance context.
- Autonomous agents that reason across steps and invoke tools.
- Evidence-grounded outputs and traceability metrics.
- Design-science plus experimental evaluation.
- A C#/ .NET prototype harness.
- Deterministic tests and LLM-assisted evaluation.
- Reproducibility through hashes and run artefacts.
- A distinction between task completion and assurance readiness.
- Industry and governance-oriented outputs.

The primary unit of analysis is whether an autonomous financial action is authorised, bounded and evidentially reconstructable.

34. Recommended Final Position

Recommended PhD title

Trustworthy Agentic Payments: A Framework for Agent Identity, Delegated Authority and Evidence Traceability in Digital Payment Systems

This title is recommended because it is internationally portable, specific enough to communicate the doctoral contribution, and broad enough to remain relevant across different payment infrastructures.

Nigeria can still be used as a principal implementation or case-study context without restricting the title geographically.

Recommended next action: take the title, research problem, primary RQ, three sub-RQs, conceptual architecture and scope boundary to the supervisor for approval before making irreversible changes to the existing PhD proposal or repository.

Appendix A. Conceptual Mapping

General autonomous-agent concept	Trustworthy agentic-payments concept
Autonomous investigative agent	Autonomous financial agent
investigative case	Transaction scenario
Investigative claim	Transaction intent / decision
Case evidence	Transaction-supporting evidence
Evidence citation	Evidence provenance reference
investigative benchmark	Agentic-payment trust evaluation harness
Assurance pass/fail	Approve / deny / escalate + trust profile
Human review of regulated-finance conclusion	Human approval of high-risk transaction
Regulatory trust	Financial accountability and auditability

Appendix B. Minimal MVP for the First Six Months

45. Create a new repository and solution.
46. Implement Agent Registry and signed agent credential.
47. Implement Principal-Agent Binding store.
48. Implement Delegated Authority JSON schema.
49. Implement deterministic Policy Engine.
50. Implement three decisions: APPROVE, DENY, ESCALATE.
51. Implement mock Payment Adapter.
52. Implement Evidence Manifest and Audit Record.
53. Create 20 ground-truth transaction scenarios.
54. Write deterministic xUnit tests.
55. Add experiment runner and JSON results.
56. Reproduce the same run using two different agent models.
57. Measure policy accuracy, evidence F1 and authorisation latency.

Appendix C. Example End-to-End Diesel Scenario

A logistics company delegates procurement authority to an autonomous agent. The agent may buy diesel from approved suppliers up to ₦50,000 per transaction and ₦200,000 per day. Human approval is required above ₦40,000.

EVENT
Fuel sensor = 22%

AGENT INTENT
Buy diesel from ABC Energy
Amount = ₦39,500

```
TRUST-LAYER CHECKS
Agent credential valid          PASS
Principal binding valid        PASS
Authority active                PASS
Purchase category allowed      PASS
Merchant approved              PASS
Per-transaction limit          PASS
Daily aggregate limit          PASS
Human approval required        NO
Evidence present                PASS

DECISION
APPROVE

PAYMENT ADAPTER
Submit controlled transaction to PSP sandbox

AUDIT PACKAGE
Agent ID
Principal ID
Authority ID
Sensor evidence
Supplier quote
Policy version
Decision reasons
Provider response
Timestamp
Evidence hash
```

Appendix D. Research Status

Source-derived foundation: the existing the earlier prototype viva presentation supplied by the researcher, including its research problem, evidence-traceability metrics, design-science methodology, prototype harness, reproducibility controls, expected contributions and work plan.

Status: the Trustworthy Agentic Payments framework, principal-agent trust model, delegated-authority representation, payment-control architecture, scenario suite and proposed metrics are presented for supervisor discussion and refinement. They should not be treated as formally approved doctoral scope until confirmed by the supervisory team.

Imagine you tell your AI agent:

“Every week, check my usual household items. If milk, bread, eggs, rice or detergent are running low, reorder them from my approved supermarket. Never spend more than £50 without asking me.”

The flow would be:

1. **You authorise the agent**
 - Allowed merchant: Tesco
 - Allowed category: groceries
 - Max automatic spend: £50
 - Anything above £50: ask you first
2. **The agent detects a need**
 - Milk low
 - Eggs low
 - Detergent finished
3. **The agent creates a payment request**
 - Tesco
 - £34.20
 - Purpose: household groceries
4. **The trust framework checks**
 - Is this the correct agent?
 - Is it acting for you?
 - Is Tesco approved?
 - Are groceries permitted?
 - Is £34.20 below the £50 limit?
 - Is there evidence showing why the purchase is needed?
5. **If everything is valid**
 - Payment is approved
 - Order is placed
6. **You can later see**
 - what the agent bought;
 - how much it spent;
 - why it bought it;
 - which permission allowed it;

- which shop received the payment.

So instead of giving an AI unrestricted access to your bank account, you give it **limited financial authority**.

Another everyday example could be your electricity bill:

“Pay my electricity bill automatically every month, but only to this provider and never more than £180.”

If the bill is £92:

APPROVE

If someone tries to make the agent send £900 somewhere else:

DENY

If the electricity bill suddenly becomes £250:

ASK HUMAN FIRST

That is the simplest way to understand your PhD idea:

AI can act for you financially, but only inside rules you gave it, and every action must be explainable and auditable.

1. Rent payment agent

You tell the agent: “Pay my rent on the 1st of every month, but only to my registered landlord and never more than £1,200.”

If the landlord details change, the agent must ask you first.

2. Fuel purchase for a car or business vehicle

The agent can pay for fuel only at approved stations and only up to a daily limit.

A £60 fuel purchase may be approved; a £600 transaction would be blocked.

3. School-fee payment

A parent authorises an agent to pay tuition fees to a specific school each term.

The agent can pay the verified school account but cannot transfer the money elsewhere.

4. Mobile-phone and internet bills

The agent checks your monthly phone and broadband bills and pays them automatically if they are within normal limits.

If your £45 bill suddenly becomes £300, it escalates to you.

5. Business supplier payments

A small shop gives an AI purchasing agent authority to reorder stock from approved wholesalers.

It can pay invoices under £1,000 but requires manager approval above that amount.

6. Restaurant stock purchasing

A restaurant agent detects that rice, cooking oil, or drinks are running low.

It orders replacements and pays approved suppliers within a weekly purchasing budget.

7. **Travel booking agent**

You tell your agent: “Book my Manchester-to-London train whenever I have a confirmed London meeting. Maximum £120.”

The agent can buy the ticket, but cannot book a £500 first-class ticket without approval.

8. **Subscription management**

Your agent can renew approved subscriptions such as Microsoft 365, Netflix, cloud hosting, or software tools.

If the renewal price increases beyond a threshold, it asks you before paying.

9. **Emergency household repair**

You allow an agent to pay approved plumbers or electricians up to £150 for urgent repairs.

If the repair quote is £450, the transaction is paused for your approval.

10. **Employee expense reimbursement**

A company's AI agent checks employee receipts against company policy.

A valid £25 taxi receipt can be reimbursed automatically, while a £700 luxury-hotel claim could be escalated or rejected.

The same pattern appears in all of them:

Need arises → AI proposes payment → identity checked → authority checked → rules checked → evidence checked → approve, deny, or ask a human.

That is what makes the concept much bigger than just “AI making payments.” It is about **controlling what financial authority an AI is allowed to exercise on someone else's behalf.**